

Disproof of conjecture 00000001110

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Verdict: FALSE

We disprove conjecture 00000001110 as stated. The refutation is unconditional and completely explicit: for the symmetric group S_6 (type A_5 , Coxeter number $h = 6$) the number of reduced words of the longest element w_0 is

$$\#\text{Red}(w_0) = 292864 = 2^{11} \cdot 11 \cdot 13,$$

and the primes 11 and 13 both exceed $h = 6$. Hence the reduced-word count is *not* h -smooth, and the conjecture fails. The smallest counterexample is $n = 6$; the cases $n = 2, 3, 4, 5$ are h -smooth.

1 The conjecture, quoted

From `conjectures/00000001110.md`:

English. Conjecture: The prime factorization of the number of reduced words of the longest element w_0 involves only primes $\leq h$ (smoothness of reduced-word counting).

The conjecture is filed in both English and Chinese. The two versions are identical in content; the Chinese original is reproduced verbatim (in Unicode) in `README.md`, because the PDF font does not embed CJK glyphs. The mathematical content is the sentence quoted above.

2 Disambiguation: what are h and the scope?

The filed statement never defines h nor says which groups are quantified over. We fix the standard reading and record it as a deliberate convention:

1. **Scope.** All finite Coxeter groups W , with w_0 the longest element and $\text{Red}(w_0)$ the set of reduced words (reduced decompositions) of w_0 with respect to the simple reflections. Under this reading, the symmetric group S_n (type A_{n-1}) is included.
2. **Meaning of h .** h is the *Coxeter number* of W . In type A_{n-1} , i.e. $W = S_n$, one has

$$h = n.$$

We stress that this is the only reading under which the statement has any content: the conjecture is a claim about a single numerical invariant h attached to the group, and the Coxeter number is that invariant.

Any universal statement over finite Coxeter groups is refuted by a single group for which the conclusion fails. We produce such a group in Section 5. Section 7 shows that the other natural readings of h do not save the conjecture either.

3 The count: a classical bijection

Theorem 1 (Staircase bijection). *For $n \geq 2$ there are bijections*

$$\text{Red}(w_0) \longleftrightarrow \text{SYT}(n-1, n-2, \dots, 1) \longleftrightarrow \mathcal{L}(\Pi_n),$$

where $\text{SYT}(\lambda)$ is the set of standard Young tableaux of shape λ , and $\mathcal{L}(\Pi_n)$ is the set of linear extensions of the staircase poset Π_n : the cells (i, j) of the staircase shape $(n-1, n-2, \dots, 1)$ with the order $(i, j) \prec (i, j+1)$ (same row) and $(i, j) \prec (i+1, j)$ (same column).

Proof sketch. The standard bijection sends a reduced word $w_0 = s_{i_1} s_{i_2} \cdots s_{i_N}$ ($N = n(n-1)/2$) to the standard Young tableau obtained by inserting k into the box of the k -th simple reflection under the Robinson–Schensted correspondence; the recording tableau of a reduced word is the staircase tableau, and this is injective onto SYT of the staircase shape. Equivalently, reading the generators s_i from left to right produces exactly the linear extensions of Π_n . The number of boxes is $N = 1 + 2 + \cdots + (n-1) = n(n-1)/2$, the number of positive roots. $\square$

We use this bijection for two things: it turns $\# \text{Red}(w_0)$ into a tableau count, to which the hook-length formula applies; and it gives an independent computational route through linear extensions of Π_n .

4 The hook-length computation for $n = 6$

For $n = 6$ the staircase shape is

$$\lambda = (5, 4, 3, 2, 1), \quad |\lambda| = 5 + 4 + 3 + 2 + 1 = 15 = \frac{6 \cdot 5}{2}.$$

The cells have hook lengths

$$\begin{array}{ccccc} 9 & 7 & 5 & 3 & 1 \\ 7 & 5 & 3 & 1 & \\ 5 & 3 & 1 & & \\ 3 & 1 & & & \\ 1 & & & & \end{array} \quad \text{i.e.} \quad \prod_{(i,j) \in \lambda} h(i,j) = 9 \cdot 7^2 \cdot 5^3 \cdot 3^4 \cdot 1^5 = 4465125.$$

In general, for the self-conjugate staircase with $m = n-1$ rows the hook length $2k-1$ occurs with multiplicity $m+1-k$, $k = 1, \dots, m$; here $m = 5$ and the multiplicities are 5, 4, 3, 2, 1.

The hook-length formula

$$f^\lambda = \frac{|\lambda|!}{\prod_{(i,j) \in \lambda} h(i,j)}$$

therefore gives

$$\# \text{Red}(w_0) = \frac{15!}{9 \cdot 7^2 \cdot 5^3 \cdot 3^4} = \frac{1307674368000}{4465125} = 292864. \quad (1)$$

Proposition 2. $292864 = 2^{11} \cdot 11 \cdot 13$, and 11, 13 are prime.

Proof. $2^{11} = 2048$, $11 \cdot 13 = 143$, and $2048 \cdot 143 = 292864$. The numbers 11 and 13 have no divisors other than 1 and themselves. Alternatively, exact trial division of 292864: it is even, and $292864/2^{11} = 143 = 11 \cdot 13$. $\square$

5 Why this refutes the conjecture

In type A_5 , i.e. $W = S_6$, the Coxeter number is $h = 6$. By Theorem 1 and (1), and by Proposition 2, the reduced-word count has prime factor 11, and

$$11 > 6 = h.$$

So the count is not h -smooth. Since the conjecture (read as standard, Section 2) asserts h -smoothness for *every* finite Coxeter group, this one group falsifies it:

Theorem 3. *The statement “ $\# \text{Red}(w_0)$ has all prime factors $\leq h$ ” is false. In $W = S_6$ (type A_5 , $h = 6$),*

$$\# \text{Red}(w_0) = 292864 = 2^{11} \cdot 11 \cdot 13, \quad 11, 13 > h = 6.$$

The smallest counterexample is $n = 6$; for $n = 2, 3, 4, 5$ the counts 1, 2, 16, 768 have all prime factors $\leq n$.

6 Counts for small n

The table lists, for $n = 2, \dots, 9$, the count of reduced words of w_0 in S_n , its exact factorisation, and the set of prime factors exceeding $h = n$. The values were computed in three mutually independent ways: the hook-length formula; a dynamic program over the weak order of S_n via $R(w) = \sum_{i: ws_i < w} R(ws_i)$, $R(e) = 1$, evaluated at w_0 ; and, for $n \leq 6$, a dynamic program over order ideals of the staircase poset. All three agree. See `reproduce.py`.

n	$h = n$	$\# \text{Red}(w_0)$ and its factorisation	primes $> h$
2	2	1	none
3	3	$2 = 2$	none
4	4	$16 = 2^4$	none
5	5	$768 = 2^8 \cdot 3$	none
6	6	$292864 = 2^{11} \cdot 11 \cdot 13$	11, 13
7	7	$1100742656 = 2^{18} \cdot 13 \cdot 17 \cdot 19$	13, 17, 19
8	8	$48608795688960 = 2^{25} \cdot 3 \cdot 5 \cdot 13 \cdot 17 \cdot 19 \cdot 23$	13, 17, 19, 23
9	9	$29258366996258488320 = 2^{34} \cdot 3 \cdot 5 \cdot 17^2 \cdot 19 \cdot 23 \cdot 29 \cdot 31$	17, 19, 23, 29, 31

The failure is not an isolated accident: every $n \geq 6$ in the table violates the bound, the smallest being $n = 6$.

7 Rescue readings, and why they fail

We tried to save the conjecture by varying h or the scope; none succeeds.

Reading	Outcome
$h = n - 1$ (the height $2h^\vee - 2$ of the highest root divided by 2, or the top degree)	at $n = 6$, still $11, 13 > 5$; refuted.
$h = n/2$	at $n = 6, 11, 13 > 3$; refuted.
Restrict to simply-laced types	S_6 (type A_5) is simply-laced, so the witness lies inside the restricted family; refuted.
Count commutation classes of reduced words instead ($s_i s_j = s_j s_i$ for $ i - j > 1$)	fails even harder: $n = 5$ gives $62 = 2 \cdot 31$ with $31 > 5$, and $n = 6$ gives $908 = 2^2 \cdot 227$ with $227 > 6$; refuted.
Restrict the rank to $n \leq 5$	the statement becomes true ($1, 2, 16, 768$ are all n -smooth), but this is a different, weaker claim than the conjecture as filed.
Use an artificial h , e.g. the number of positive roots $N = n(n - 1)/2 = 15$ at $n = 6$	at $n = 6$ this covers $11, 13$; but $15 \neq h$ is not the Coxeter number and the reading is not the standard one. Recorded as a caveat, not as a rescue.

The only readings that avoid the counterexample change the meaning of h away from the Coxeter number, or restrict the rank to $n \leq 5$. Under the standard reading (Coxeter number, all finite types, type A included) the conjecture is false, with the smallest witness $n = 6$.

8 Caveats

- **The file does not pin down h or the scope.** The refutation adopts the standard reading (Section 2): h is the Coxeter number and the scope is all finite Coxeter groups, so type A_{n-1} with $h = n$ is included. If one instead restricts the scope to $n \leq 5$ or redefines h artificially, the statement can be made true; this is stated openly in Section 7. It is a change of the claim, not a rescue of the filed one.
- **The bijection is cited, not formalised.** The classical equivalence “reduced words of w_0 in $S_n \leftrightarrow$ standard Young tableaux of the staircase shape $(n - 1, \dots, 1) \leftrightarrow$ linear extensions of the staircase poset” is used as a known theorem (Theorem 1); core Lean, in which the accompanying formalisation is written, has no Coxeter groups, so the Lean development formalises the arithmetic evaluation of the hook-length formula rather than the bijection itself. The count is nevertheless confirmed independently by two dynamic programs in `reproduce.py`.
- **The count is a *number of reduced words*, not the number of commutation classes.** The conjecture says “number of reduced words”; we take it literally. Counting commutation classes instead is a different quantity and fails even more quickly, as recorded above.
- **Only finite types are considered**, since w_0 and the Coxeter number exist in that setting; this is the natural scope of the statement.

9 Reproducibility and formalisation

`reproduce.py` (Python 3 standard library only, no `sympy`) computes the counts for $n = 2, \dots, 9$ by the hook-length formula and by the weak-order dynamic program (plus an order-ideal dynamic program for $n \leq 6$), factorises them with a self-written trial-division routine, prints the table with

the set of primes exceeding n , asserts that $n = 6$ violates the bound with $11, 13 > 6$, identifies $n = 6$ as the smallest violation, and exits 0 on PASS:

```
$ python3 reproduce.py
...
PASS: all checks verified; conjecture 00000001110 is FALSE.
```

LaTeX: `tectonic --outdir build main.tex` produces `build/main.pdf`.

The Lean 4 project under `lean4/` (core Lean, `import Std`, no `Mathlib`, no `sorry`) pins `leanprover/lean4:v4.33.1` and proves, with *no* use of `native_decide` and therefore with no `Lean.ofReduceBool` in the axiom footprint,

```
theorem reducedWordsWOS6_eq : reducedWordsWOS6 = 292864
theorem reducedWordsWOS6_factorisation :
  reducedWordsWOS6 = 2 ^ 11 * 11 * 13
theorem eleven_dvd : 11 | reducedWordsWOS6
theorem thirteen_dvd : 13 | reducedWordsWOS6
theorem not_six_smooth : ¬ Bsmooth 6 reducedWordsWOS6
```

where `reducedWordsWOS6` is $15!/(9 \cdot 7^2 \cdot 5^3 \cdot 3^4)$ computed inside Lean, and `Bsmooth B c` means $\forall p, p \mid c \rightarrow p \leq B$. The audit `Check.lean` shows every one of these theorems depends on *no axioms at all*.

Status against the submission rules

- **LaTeX source** — `main.tex`, a standalone article using `geometry`, `amsmath`, `amssymb`, `amsthm`, `array`, `parskip`, `booktabs`; compiles with `tectonic main.tex`.
- **PDF** — at `build/main.pdf`, produced by `tectonic --outdir build main.tex`.
- **Lean 4 project** — `lean4/` with `lean-toolchain` pinned to `leanprover/lean4:v4.33.1`, `lakefile.toml` (library `Main`, project `tlmc1110`, no dependencies), `Main.lean`, `Check.lean` (`#print axioms`), and `lean4/README.md`. `lake build` exits 0; the audit reports neither `sorryAx` nor `Lean.ofReduceBool` (in fact no axioms at all).